

HYDRAULISCHE ARBEITSGERÄTE DAVID

MATRA

Operating Instructions

Hydraulic Power Units

DAVID

MATRA

Maschinen in aller Welt

MATRA-WERKE GmbH
Dieselstr. 30-40 · 60314 Frankfurt/M.
Postfach 111 351 · 60048 Frankfurt/M.
Telefon (0 69) 40 16-0 · Telex 4 17 118
Telefax (0 69) 4 01 61 98

Table of Contents

		Page
1	General remarks	1
2	DAVID pumps	2
3	Single-stage hand pumps	3, 4
4	Two-stage hand pumps	5-8
5	Foot (pedal) controls	9
6	Hydraulic fluid	10
7	High-pressure hoses	10
8	Rams	11
9	Couplings	12
10	Pressure gauges	13
11	Fittings	14
12	Appendix: Spare part drawings	15

Notes on Units of Measurement

In accordance with the "law on units of measurements" which came into force (in Germany) on July 2, 1970, and to German Standard DIN 1301, only the following units may now be used in this field:

Unit of force	Newton = N =	1 kgm/s^2
Unit of pressure	Bar = bar =	10 N/cm^2

Since introduction of these units will take some time, both old and new designations will remain in use during the transitional period due to end on December 31, 1977.

The following factors will be helpful in converting old into new units:

Power	1 N = 1 kgm/s^2	
	10 N = 1.02 kp	May be rounded off:
	9.81 N = 1 kp	10 N = 1 kp
	1,000 N = 1 kN	10 kN = 1 Mp

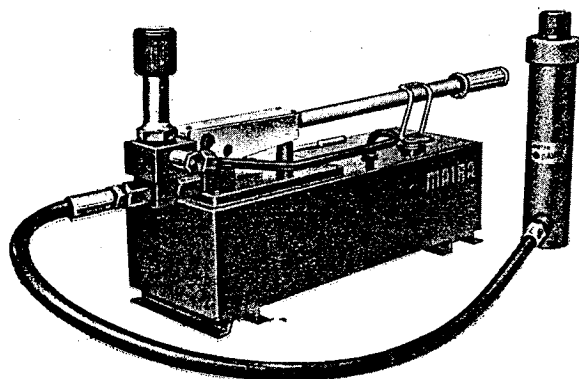
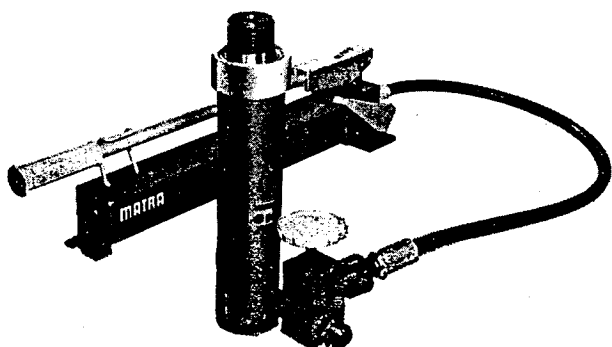
Pressure	1 bar = 10 N/cm^2	
	1 bar = 1.02 kp/cm ²	May be rounded off:
		1 bar = 1 kp/cm ²

This operating manual remains our property. Unauthorised reproduction, in whole or in parts, utilization or communication to third parties, particularly competing companies, is prohibited.

(Copyright law of 19.6.1901 and UWG of 7.6.1909).

1. General Remarks

The basic set of DAVID hydraulic power units is always made up of one pump (hand, foot or electric operation) and one ram (press, pull or push/pull model, special power units, two-column press, etc.) connected by one or more hydraulic hoses. Pumps and rams are arranged for a maximum service pressure of 625 bar.

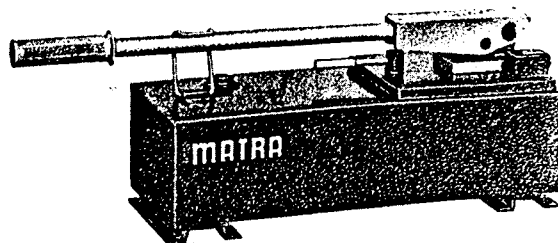
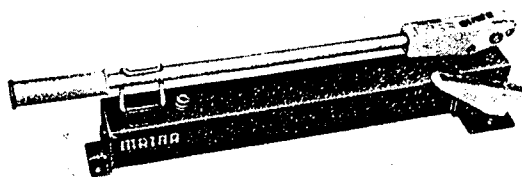


Irrespective of pump location, rams can be applied to the workpiece, or built into fixtures or machines, in any desired position. Most rams are equipped with mounting thread on the cylinder. For rams of 60, 100, 160, 200 and 400 kN (6, 10, 16, 20 and 40 Mp), press frames are supplied, so that they can be assembled into complete hydraulic presses in the simplest way. Other special attachments and a wide range of accessories convert the rams into power units of universal application. A variety of fittings further extend the range of duties.

For dimensions and technical data of DAVID units, consult our comprehensive DAVID catalogue or the separate leaflets.

No special technical knowledge or skills are needed for working with DAVID hydraulic power units. But you should study and carefully observe the following instructions to avoid operating troubles and make it easier to remedy any faults that might occur.

In the standard version, DAVID units are designed for general workshop duties. When used for special duties, a higher degree of wear will be unavoidable, so that it is advisable in such cases to select a special-purpose version. Service temperatures should not lie below -15°C and must not exceed $+80^{\circ}\text{C}$. Where higher temperatures are unavoidable, special seals may have to be provided. Power units should be kept scrupulously clean and protected against all damage, particularly from coarse dirt particles, dust, sand and similar abrasive matter, damp and abnormal climatic conditions. Bearings and sliding parts should be regularly lubricated.



2. "DAVID" Pumps

DAVID hydraulic power units are operated by pumps of various sizes, hand, foot or electrically controlled. Pumps for dual-acting rams (push/pull versions) are fitted with a special change-over valve to direct the oil flow as required.

Pumps are connected to the various rams by high-pressure hose and coupling. Selection of pump and ram size is governed by effective pump oil volume which must be at least equal to oil consumption (displacement per stroke) of the connected ram. For rams with dual-acting piston, consumption is calculated from the difference between displacement per stroke for the pull and push action, since oil is returned to the pump during the piston traverse.

Pumps are fitted with a safety valve, set as standard to the specified service pressure of 625 bar, so preventing any overrun beyond permissible limit on the pump side. This valve setting must not be tampered with, otherwise pump and connected power unit may sustain damage.

Summary of DAVID hand pump models

Single-stage hand pump

to operate rams with single-acting piston (press, pull, etc.)

	No. 4601.00000.330
Oil capacity	900 cc
Effective oil volume	730 cc
Delivery per pump stroke	2, 5 cc

Two-stage hand pumps

to operate rams with single-acting piston (press, etc.)

	No. 4601.00000.360	No. 4601.00000.380
Oil capacity	3500/5000 cc	8000 cc
Effective oil volume	2700/4200 cc	6800 cc
Delivery per pump stroke	12/2, 5 cc	12/2, 5 cc

Two-stage hand pumps

to operate rams with dual-acting piston (push/pull)

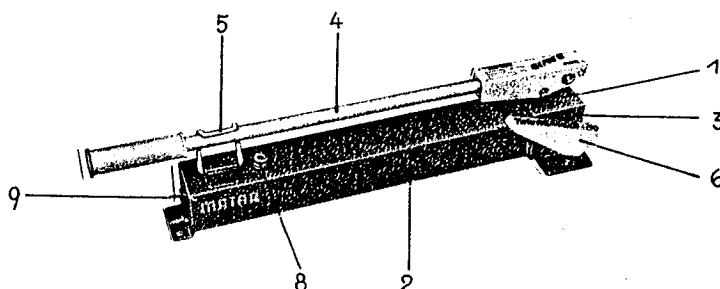
	No. 4601.00000.850	No. 4601.00000.890
Oil capacity	3500/5000 cc	8000 cc
Effective oil volume	2700/4200 cc	6800 cc
Delivery per pump stroke	12/2, 5 cc	12/2, 5 cc

Separate operating manual is supplied for electrically controlled DAVID pumps.

The following remarks apply to all pump models:

1. Pumps must always be filled with an adequate quantity of suitable hydraulic fluid.
2. Pumps should not be filled with more than the specified volume, otherwise the overpressure set up in the oil tank by a forceful piston return may lead to damage or even fracture.
3. If excessive underpressure in the oil tank inhibits suction intake, open the filler plug to correct.
4. Should the ram not retract properly, also open the filler plug and check carefully that the ram correctly returns to its starting position. Close the plug only when the ram has reached its end position, otherwise too much air will be trapped in the tank.
5. To drive out any air bubbles trapped in the compression chamber, hold the pump, body facing down, open the return valve, then smartly move the pump lever to and fro several times.
6. Any foreign bodies deposited on the valve seats can be flushed away by repeated fast pumping strokes, at the same time alternately opening and closing the return valve or shifting the lever of the change-over valve to and fro.
7. Always operate return and change-over valves slowly and smoothly to prevent foaming of the hydraulic fluid.

3. Single-stage Hand Pump, model 4601.00000.330



This pump is designed for operation of power units with single-acting ram. It is equipped with a single-stage submersible piston; pump body (1) and oil tank (2) form an integral unit and cannot be separated.

Normally, the pump works in the horizontal position, but it can also be located vertically if required, when pump body (1) must face down to ensure that the intake is always covered by the oil level.

Pressure port (3) for the high-pressure hose or required fittings is located at the front face of body (1). It is tapped for M 14 x 1.5 - 12 mm deep and sealed at the thread base. Pump piston and return valve are actuated by lever. A force of up to approx. 45 kp must be applied at the handle of lever (4) to build up the service pressure. Lever (4) can be locked in the horizontal position with spring clip (5) and so serves as carrying handle. To open the return valve, press down lever 6. The lever can be fitted into a second position on the valve spindle to compensate for any valve seat wear.

Place the pump horizontally to fill in the oil. Take off filler plug (8), pour in the oil through the inlet on top of the tank. A built-in pipe automatically provides the necessary air cushion, so that you can safely fill the pump completely (900 cc).

Use only clean oil in the pump. From time to time, filter the oil and at the same time flush out the tank. Take off flushing plug (9) from the end of tank (2), pour in the rinsing fluid through inlet (8) and let it drain from port (9).

We supply the oil in 1-litre containers, sufficient for one filling.

Keep the pump clean at all times and protect it from damage, particularly by coarse dirt particles, dust, sand, etc., damp and abnormal climatic conditions. Regularly lubricate bearings and all sliding parts.

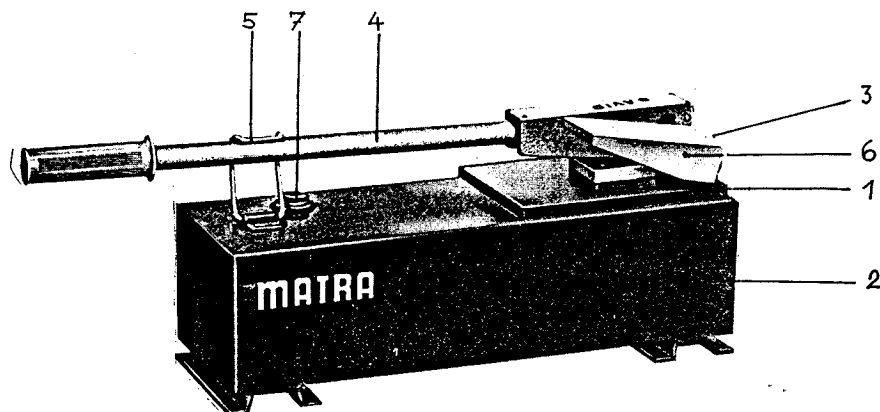
Notes on possible troubles and their remedy will be found overleaf.

Single-stage Hand Pump, model 4601.00000.330

Troubles, their Causes and Remedies

<u>Fault</u>	<u>Cause</u>	<u>Remedy</u>
Insufficient delivery	Insufficient oil	Top up
	Underpressure in tank	Briefly open filler plug
	Air trapped in pump compression chamber	Hold pump vertically (body facing down), pump several times in quick succession, alternately opening and closing the return valve.
Ram does not properly retract	Foreign body on valve seats prevents valves closing all the way	As above: if necessary, clean valve balls and seats
	Too much oil	Reduce oil volume
	With ram wholly or partly advanced, air has entered tank (e.g. during topping up), too much air in tank for return travel, so that all the oil cannot return.	Take off filler plug (8), re-fit it only when ram has fully returned.
Pressure drops	Foreign body on valve seats	As above under "insufficient delivery".

4. Two-stage Hand Pumps, model 4601.00000.360, model 4601.00000.370



The two models have the same pump (1), but tanks (2) of different size and are designed for operation of single-acting rams. Pressure port (3) is closed off by tightly sealing blanking plug as standard. Remove this plug to couple up the high-pressure hose. The port is tapped for M 14 x 1.5 - 12 mm deep and sealed at the thread base. Oil flow is controlled by the return valve. Press down valve lever (6) to open it, press it up to close the valve.

The two delivery stages produced by a stepped piston give an automatic change-over from low to high pressure when a service pressure of approx 100 bar has built up. In the low pressure range both delivery stages are in action, in the high pressure range only the low stage. Pump (1) delivers 12 cc/stroke in the low pressure, 2.5 cc/stroke in the high pressure range. The force to be applied to the handle of pump lever (4) is approx. 8 kp at the change-over limit, approx. 45 kp to reach service pressure. Lever (4) can be locked in the horizontal position with spring clip (5) to serve as carrying handle.

Place the pump horizontally to fill in the oil. Take off filler plug (7) and pour in the oil through the inlet. A built-in pipe automatically provides the necessary air cushion, so that you can safely fill the pumps completely.

Oil capacities: Model 4601.00000.360 = 5000 cc
 Model 4601.00000.370 und 4601.00000.380 = 8000 cc

Use only clean oil in the pump, filter it from time to time and flush out the tank.

We supply oil in containers holding one filling.

Keep the pumps clean at all times and protect them from damage, particularly by coarse dirt particles, dust, sand, etc. damp and abnormal climatic conditions. Regularly lubricate bearings and all sliding parts.

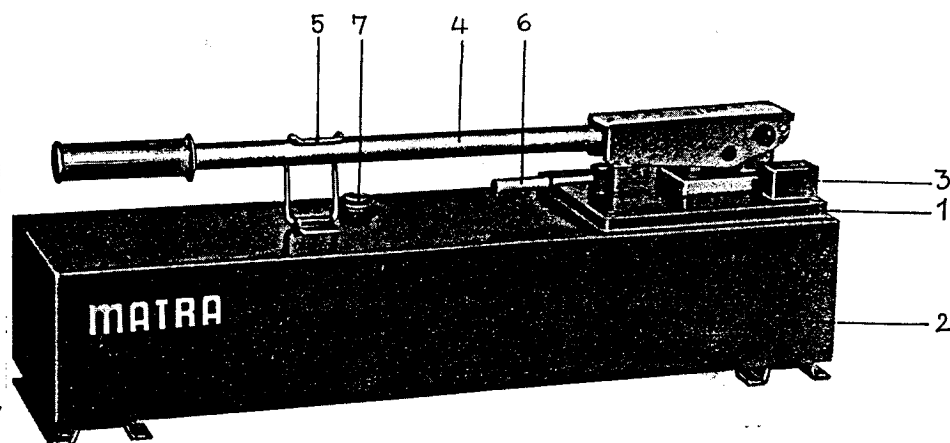
Notes on possible troubles and their remedy will be found overleaf.

Two-stage Hand Pumps, model 4601.00000.360, model 4601.00000.370

Troubles, their Causes and Remedies

<u>Fault</u>	<u>Cause</u>	<u>Remedy</u>
Insufficient delivery	Insufficient oil	Top up
	Underpressure in tank	Briefly open filler plug
	Air trapped in pump compression chamber	Hold pump vertically (body facing down), pump several times in quick succession, alternately opening and closing the return valve.
	Intake mesh filter dirty	Take pump (1) (fixed in tank (2) with 4 cyl. bolts) from tank and clean filter, flush out tank and filter the oil.
Ram does not retract all the way	Foreign body on valve seats prevents valves closing all the way.	With ram unloaded, pump smartly several times in all positions of lever (6) to flush out the dirt.
	Too much oil filled in with ram fully moved out	Drain off surplus oil (from port (7), run filler plug down tightly only when ram has retracted completely.
	Trapped air causes overpressure in tank	Take off filler plug, re-fit it only when ram has retracted all the way.
Pressure drops	Foreign body on valve seats	Smartly move pump lever in all positions of the change-over valve, when pump should repeatedly change from low to high pressure.

Two-stage Hand Pump, model 4601.00000.850, model 4601.00000.890



Both models, sharing the same pump (1), but having tanks (2) of different size, are designed for operation of single-acting and dual-acting rams. Pressure ports (3) are closed off as standard with blanking plugs giving a tight seal. To work single-acting rams off the pump, take off one blanking plug. To connect one dual-acting ram or two independently working single-acting rams, take off both plugs. The oil flow is controlled by the lever of change-over valve (6) under pump lever (4). The lever has three positions. When working with single-acting rams, it points to the pressurized port when set to ram advance, to the closed port when set to ram retraction, or it may stand at centre.

When working with dual-acting rams, lever (6) points to whichever port is pressurized when set to ram advance, while the other port is unpressurized and set to return. In the centre position, both ports are unpressurized. Ports are tapped for M 14 x 1.5 - 12 mm deep and sealed at the thread base.

The stepped piston of pump (1) produces the two pressure stages, the pump changing automatically from low to high pressure as soon as a service pressure of approx. 100 bar has built up. In the low pressure range, both delivery stages are in action, in the high pressure stage only the lower stage. Pump (1) delivers 12 cc/stroke at low pressure, 2.5 cc/stroke at high pressure. The force to be applied to the handle of pump lever (4) at the change-over limit is approx. 8 kp, that to build up the service pressure approx. 45 kp. Lever (4) can be locked with clip (5) in the horizontal position to serve as carrying handle.

Place the pump horizontally to fill in the oil. Take off filler plug (7) and pour in the oil through the inlet. A built-in pipe automatically provides the necessary air cushion, so that you can safely fill the pumps completely.

Oil capacities: Model 4601.00000.850 = 5000 cc
 Model 4601.00000.890 = 8000 cc

Fill pumps only with clean oil; filter the oil from time to time and at the same time clean the tank.

We supply oil in containers holding one filling.

Keep the pumps clean at all times and protect them from damage, particularly by coarse dirt particles, dust, sand, etc., damp and abnormal climatic conditions. Regularly lubricate bearings and all sliding parts.

Notes on possible troubles and their remedy will be found overleaf.

Two-stage Hand Pumps, model 4601.00000.850, model 4601.00000.890

Troubles, their Causes and Remedies

<u>Fault</u>	<u>Cause</u>	<u>Remedy</u>
Insufficient delivery	insufficient oil	Top up
	Underpressure in tank	Briefly open filler plug
	Air trapped in pump compression chamber	Hold pump vertically (body facing down), pump several times in quick succession, alternately opening and closing the return valve.
	Intake mesh filter dirty	Take pump (1) (fixed in tank (2) with 4 cyl. bolts) from tank and clean filter, flush out tank and filter the oil.
Ram does not retract all the way	Foreign body on valve seats prevents valves closing all the way	With ram unloaded, pump smartly several times in all positions of lever (6) to flush out the dirt.
	Too much oil filled in with ram fully moved out	Drain off surplus oil (from port (7), run filler plug down tightly only when ram has retracted completely.
	Trapped air causes overpressure in tank	Take off filler plug, re-fit it only when ram has retracted all the way.
Pressure drops	Foreign body on valve seats	Smartly move pump lever in all positions of the change-over valve, when pump should repeatedly change from low to high pressure

5. Foot (pedal) Controls

Pedal control No. 4601.51000.330

for hand pump 4601.00000.330

Assembly Instructions

1. Mounting the pedal:

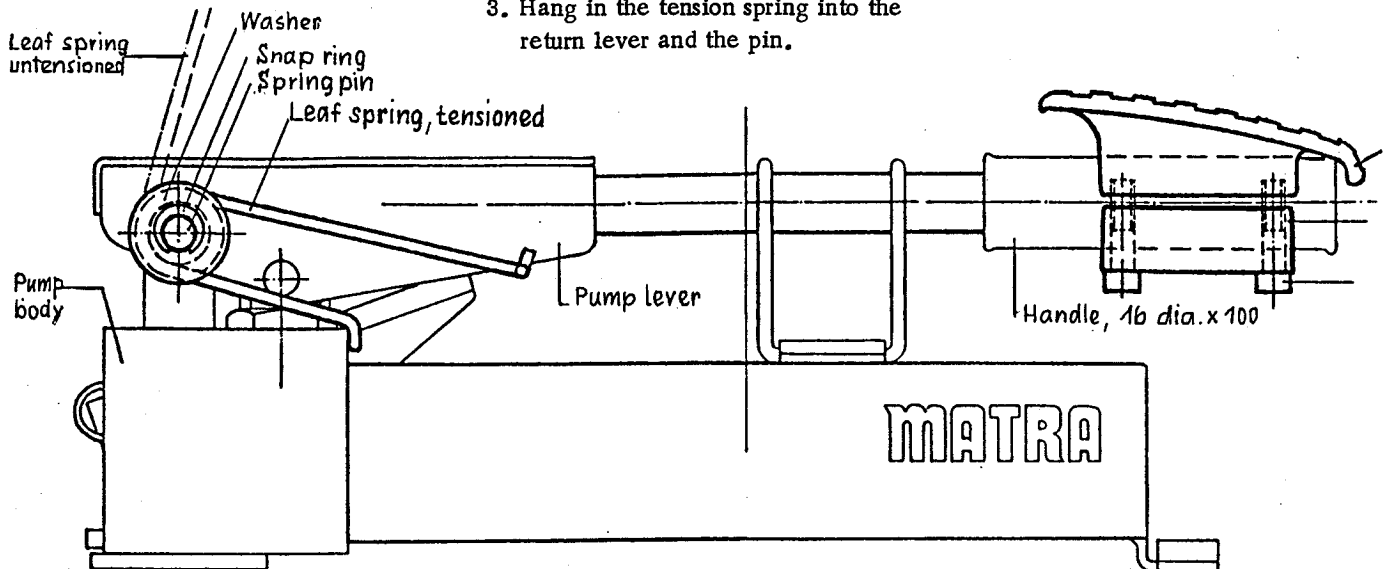
Place upper pedal section on handle, bolt to lower section with four cylindrical screws, M 6 x 30, running screws down tightly.

2. Fitting the leaf spring:

Take snap ring off rear pin on pump lever, then take off pin (not needed for pedal control), fit leaf spring over pump lever and place it on pump body. Slide spring pin No. 4601.51002.330 into free, rear bore.

Swing pump lever up, hook both spring ends into the lugs on the pump lever to tension the spring. Then fit washers, 8.4 x 28 x 2, on both sides against the spring pin and secure with snap rings 7 DIN 6799.

3. Hang in the tension spring into the return lever and the pin.



Pedal control No. 4601.51000.360

for two-stage hand pump 4601.51000.360

1. Mounting the pedal:

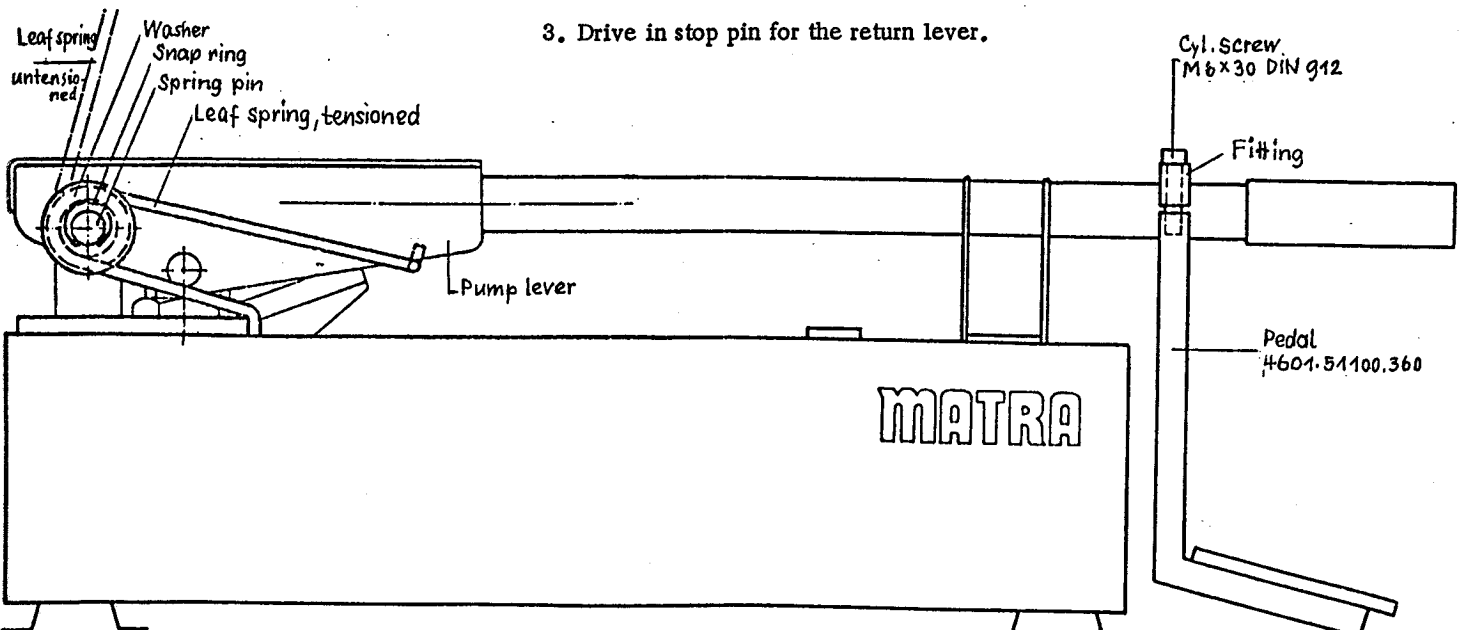
Fit lower pedal section to handle and bolt to clip with 2 cylindrical screws, M 6 x 30, running screws down tightly.

2. Fitting the leaf spring:

Take snap ring off rear pin on pump lever, then take off pin (not needed for pedal control), fit leaf spring over pump lever and place it on pump body. Slide spring pin No. 4601.51002.360 into free, rear bore.

Swing pump lever up, hook both spring ends into the lugs on the pump lever to tension the spring. Then fit washers, 8.4 x 28 x 2, on both sides against the spring pin and fit washer with flat on the pressure-port side.

3. Drive in stop pin for the return lever.



6. Hydraulic Fluid

Use only one of the hydraulic oils listed below to operate DAVID power units.

These oils have a viscosity of 3⁰ to 5⁰ E at 20⁰ C and permit working with hand pumps at temperatures down to approx. 15⁰ C below zero. Bear in mind that at temperatures below freezing point, delivery drops to some extent, as the oil thickens and becomes more difficult to draw in. But the specified high pressure is still attained.

Use of any other oils may cause swelling of the sealing cuffs and lead to operating troubles. Wholly unsuitable are: brake fluid, shock absorber oils, diesel oil, kerosene, petrol and similar fluids.

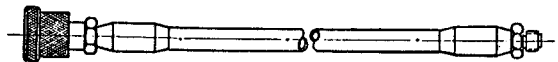
Fill pumps only with clean oil. Filter oil from time to time, flush out and clean the tanks at the same time. Top up the oil as required.

We supply the oil in containers holding one tank filling.

Low-flammability hydraulic fluids may only be used if pumps and power units are specially adapted for these. They require, for instance, the fitting of suitable seals and removal of all paint coats on contact with the fluid.

The brand of a suitable oil with all oil companies is as follows:

hydraulic oil 'H-LP 10'(ISO)

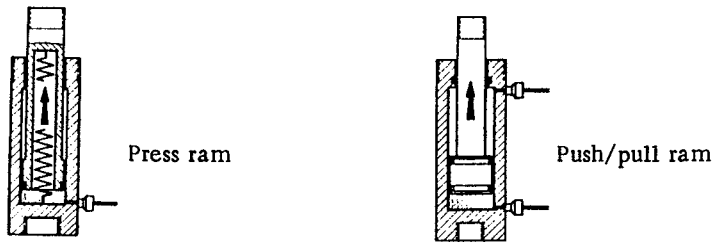


7. High-pressure Hoses

Pump and ram are connected by one or more high-pressure hoses.

Hoses have external threads, M 14 x 1.5/14 mm deep, for screwing into the pump port on one side, on the other a male coupler to fit the female coupler on the ram. Hoses are designed for a service pressure of 625 bar. Although highly resistant and tough, they should be protected, like all such hoses, from overload and damage. In particular, avoid sharp kinks, never drive over hoses and keep them away from sources of heat, e.g. welding flame.

8. R a m s



Hydraulic rams are divided into:

1. Single-acting rams (press, pull action, etc.) with 1 pressure port.
2. Dual-acting rams (push/pull action) with 2 adjoining ports, the top one leading to the push, the bottom to the pull side of the ram.

The threads on cylinder and ram are designed to take accessories and tools, the cylinder thread also serves for screwing the unit into fixtures.

Always fit a cap or similar to the ram before applying it against the workpiece under pressure and check that this is firmly run down.

It is advisable to protect external cylinder threads with a nut when not in use.

Do not continue pumping after the ram reaches its end position to prevent damage to the face or workpiece. On most rams, the end of the stroke travel is indicated by a groove marked in red.

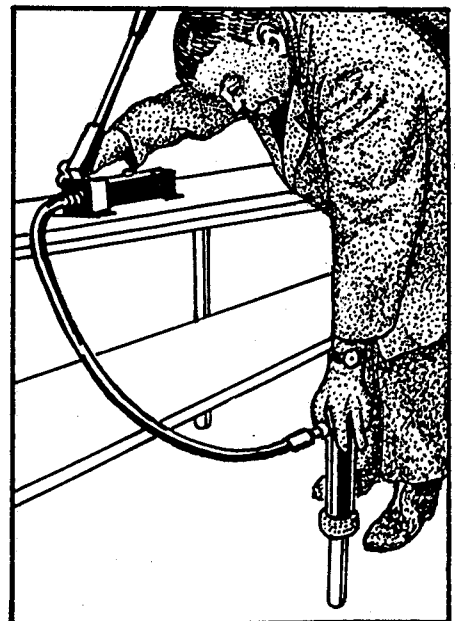
To put the power unit into service, couple the hose to the coupling half on the ram. Ram parts are tapped M 16 x 1.5/12 mm deep and sealed at the thread base. To connect ram and pump by pipe instead of hose, unscrew the coupler from the ram and replace it by pipe adaptor fitting.

Do not put hydraulic power units under eccentric or bending load to avoid damage. Take suitable measures to eliminate any possibility of such stresses, e.g. by mounting the ram in swivel suspension. Do not exceed maximum specified working pressure either on the pump or on the ram side.

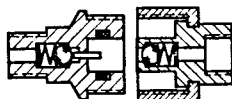
DAVID power units in the standard version are designed for general workshop duties and arranged accordingly. Their use for special duties is liable to increase wear, so that it is advisable to select a special-purpose version in such cases (e.g. with hard-chromed piston, dirt scrapers, etc.).

Keep rams clean at all times and protect them from damage, particularly by coarse dirt particles, dust, sand etc., damp and abnormal climatic conditions. Regularly lubricate all moving parts.

Before putting a power unit into service, it should be bled; as a rule, no further bleeding will be needed. Place the ram on a level lower than the pump, pressure port facing up, then pump to advance and retract the ram and so drive all trapped air bubbles up into the pump. Then take off the pump filler plug to let the surplus air escape.



9. Couplings



Uncoupled

Female coupler for ram



Coupled

Male coupler for hose

Rams are connected to hoses by coupling, having a female coupler on the ram side (M 16 x 1.5/14 mm long, external) and a male coupler on the high-pressure hose (M 14 x 1.5/12 mm deep, internal).

No tools are needed to couple up and uncouple ram and hose. Merely run down the knurled locknut by hand, leaving a little length play, so that the coupling can still just be turned by hand.

Valve balls built into the coupling automatically seal off the two halves when uncoupled and prevent loss of oil. As the coupling is locked, the balls press on each other and so open the passage to the oil flow in both directions. In other words, to open the fluid circuit, you need 2 complete coupling halves in perfect condition.

In the latest coupling version, the plunger pins actuating the opening and closing of the flow passage are set to a fixed travel.

Earlier coupling versions have an adjustable pin stroke. Particularly where new and old versions are used in combination, note the following points:

If the stroke is set too short, a sudden pressure surge may cause the balls to wobble and close off the flow to one side

If the return flow is shut off on push/pull rams, any load applied on the pressure side may set up impermissible overpressure inside the cylinder and makes it unserviceable.

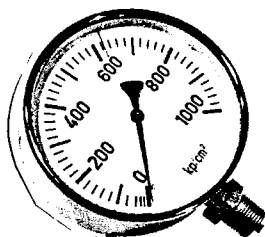
Imprecise setting or inadvertent re-setting of one coupling half during cleaning may, therefore, give rise to the troubles listed below, but these are easy to remedy:

1. The pump builds up pressure, but the ram either does not move or stops after a short travel. This is due to the ball on the hose coupler being forced by the oil flow against its seat, so that the valve is closed and does not allow oil to flow to the ram. Unscrew the coupler on the power unit and turn the slotted screw inside it clockwise to set a valve ball travel of about 1.5 mm. This will ensure that the valve opens. After re-setting, firmly screw the coupler back into the power ram.
2. The ram does not retract, even though the return valve is open and there is neither too much air nor oil in the pump. This is due to the return oil driving the ball in the ram coupler against its seat, so barring the return passage of oil to the pump. To remedy the fault, take the hose coupler off, turn the slotted screw in it clockwise to set a ball travel of approx. 1.5 mm. Finally firmly re-fit the coupler to the hose. In case of doubt, or if the remedy fails, replace the couplings. The earlier coupling version with adjustable ball travel can be recognised by the slotted screw visible from the coupling thread side.

Couplings not in use should be blanked off with the caps supplied to keep out dirt.

On stationary installations, flow resistance can be reduced (to improve return flow) by taking the internal parts out of the couplings. But in all cases, always connect by either complete couplers or couplings on which the internal parts have been removed on both sides.

10. Pressure Gauges



The gauges to register push or pull forces of hydraulic power units should be fitted into the pressure line. Though the gauges measure the fluid pressure, i.e. pump service pressure, the dials are calibrated to show directly the push or pull power applied by the ram, instead of the fluid pressure. Consequently, readings will only be correct if the gauges are used with rams specified for them.

All our DAVID power units work at a standard oil pressure which produces the working power marked on the ram.

Irrespective of the gauge model fitted, the power marked on the ram has built up as soon as the pointer swings to the red mark (maximum permissible pressure), the only exception being gauges designed for special applications.

The pressure gauge No. 4605.00000.521 has a dial which indicates the service pressure of the pump of 0 to 625 bar (see illustration). When running simultaneously different rams this pressure gauge can also be used for push/pull rams. Then the power is to be converted accordingly.

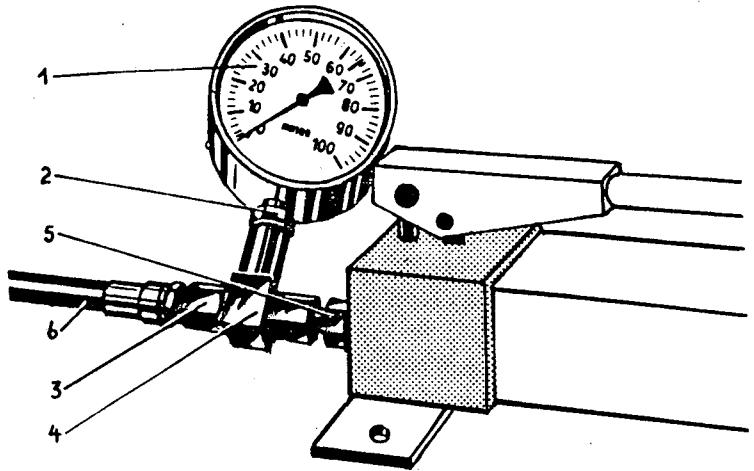
Reading tolerance of these gauges is ± 1 to 2% of full swing, working range lies at approx. 2/3rd of full swing. Precision gauges with a reading tolerance of $\pm 0.5\%$ of full swing are available on request. The instruments have a diameter of 130 mm and R 1/2" connecting thread.

A restrictor built into the connecting fitting slows down pointer swing to zero in case of sudden pressure release to prevent damage to the instrument. The restrictor does not affect pressure rise which can be read off without delay.

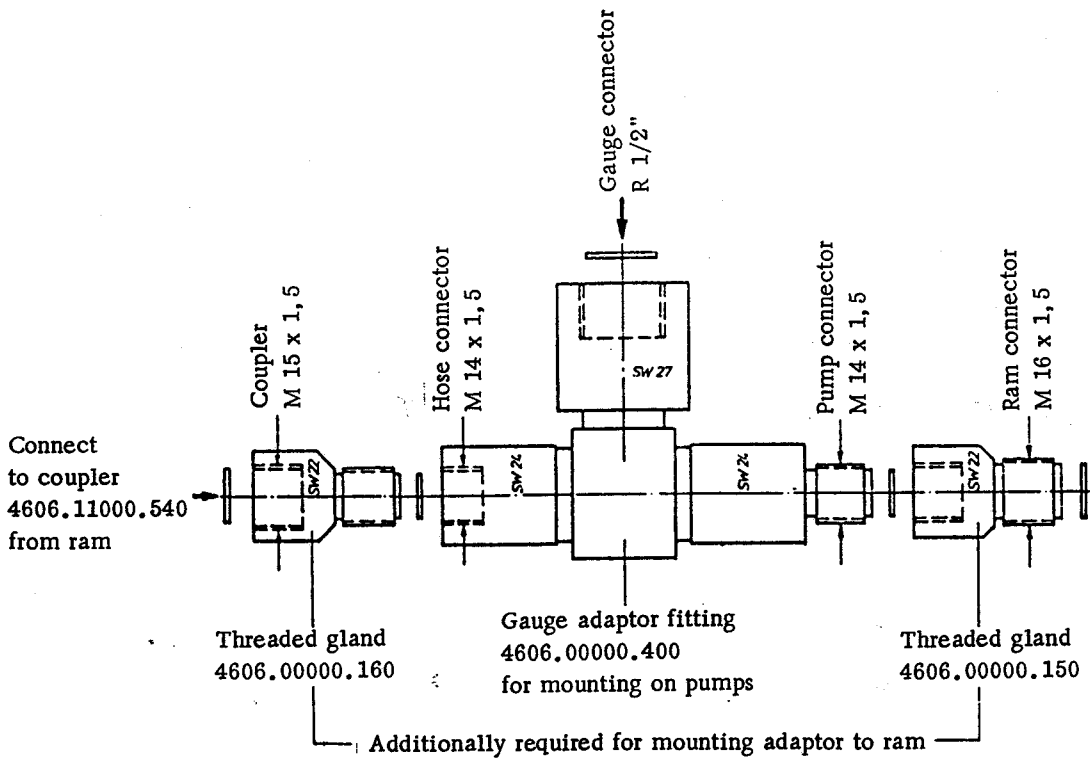
Notes on pressure gauge installation will be found overleaf.

Fitting the pressure gauges

In place of high-pressure hose (6), screw connector 4606.00000.400 to pump connector socket (5), then screw the high-pressure hose to union (3). Slacken hexagon nut (2) and union (3) to turn and tilt gauge (1) into the desired position. Then run down nut (2) and union (3) again; note that hexagon nut (2) has a right-hand thread for the gauge socket and a left-hand thread for square adaptor (4). Check that sealing rings are correctly seated. Finally, slacken hollow screw (7) and swing socket (5) to bring the whole assembly into its final location.



Connector fitting 4606.00000.400



Fittings

A wide range of special fittings is available for DAVID units. These are described in the DAVID catalogue.

For fittings designed to operate several rams a single pump, note that automatic uniform traverse of all connected rams cannot be ensured.

Without extra auxiliary controls, hydraulic drive systems cannot provide such uniform travel. The ram under the lightest load will inevitably advance first and that under heaviest load last; vice versa, the most heavily loaded ram will retract first. Additional guides are, therefore, required which must be provided on the machine parts to be moved in the form of accurate slideways, etc. which will enforce uniform movement of machine components. Alternatively, each ram can be supplied from an independent oil circuit, or the oil flow may be regulated by the shut-off valves acting as restrictors.

DAVID

Spare parts list

Spare part orders to

Matra - Werke GmbH
6 Frankfurt/M - 11
Dieselstraße 30 - 40
POB 111351
Phone : (069) 4016-0
Telex : 417118
Telefax : 069 / 4 01 61 98

Please quote part numbers in full.

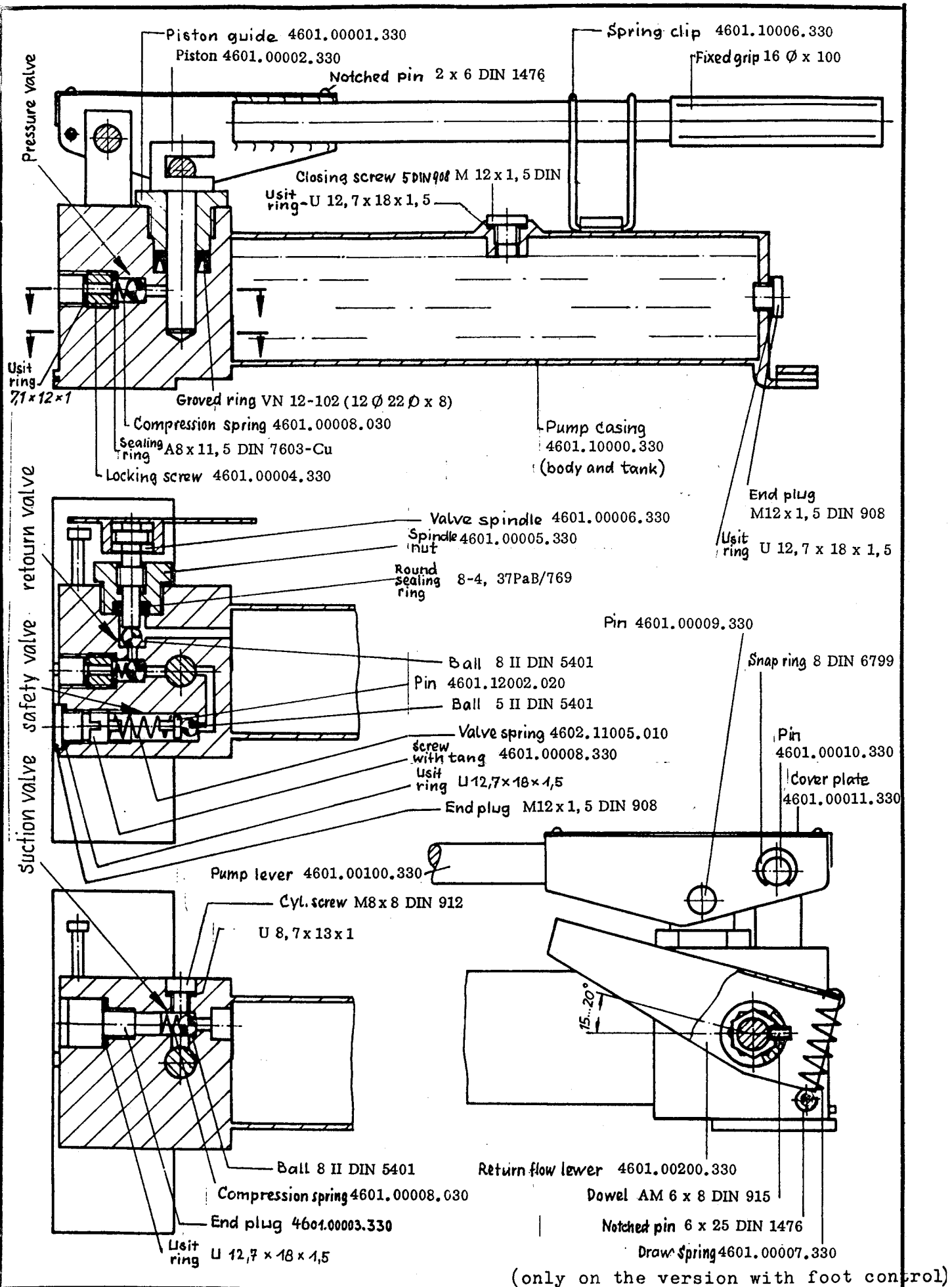
Sample parts are returned only on explicit request.

In your orders, please state desired method of despatch and accurate address for consignment.

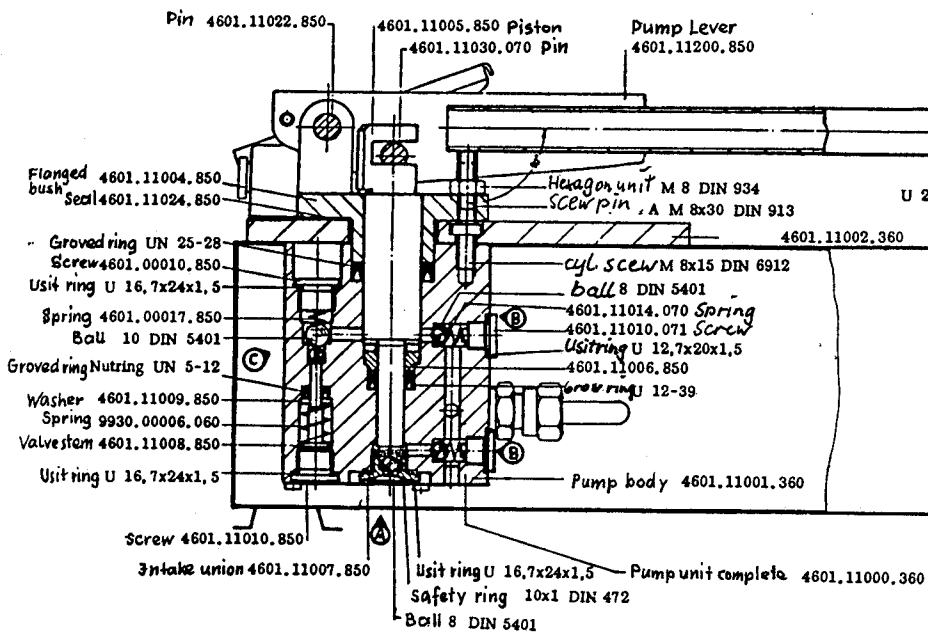
Repairs to

Matra - Werke GmbH
60314 Frankfurt/M.
Dieselstraße 30 - 40
Freight - Frankfurt/M - Ost
Express - Frankfurt/M - Hauptbahnhof

Please attach tie-on label to parts sent in.
Also please state kind of defect occurred.



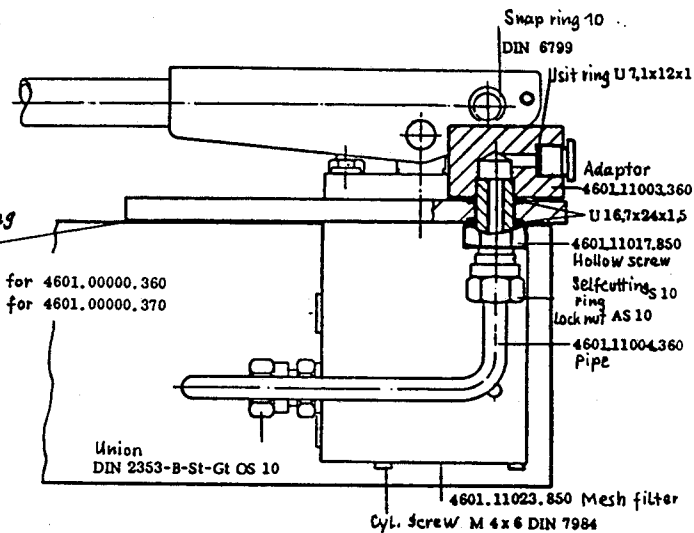
Datum 4.8.71	Spare part drawing Hand pump 4601.00000.330	Zchn.-Nr. 4601.02002.330
Name BOP		



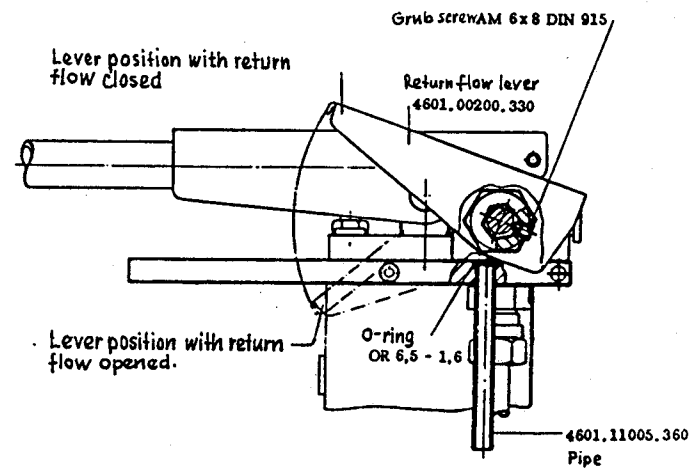
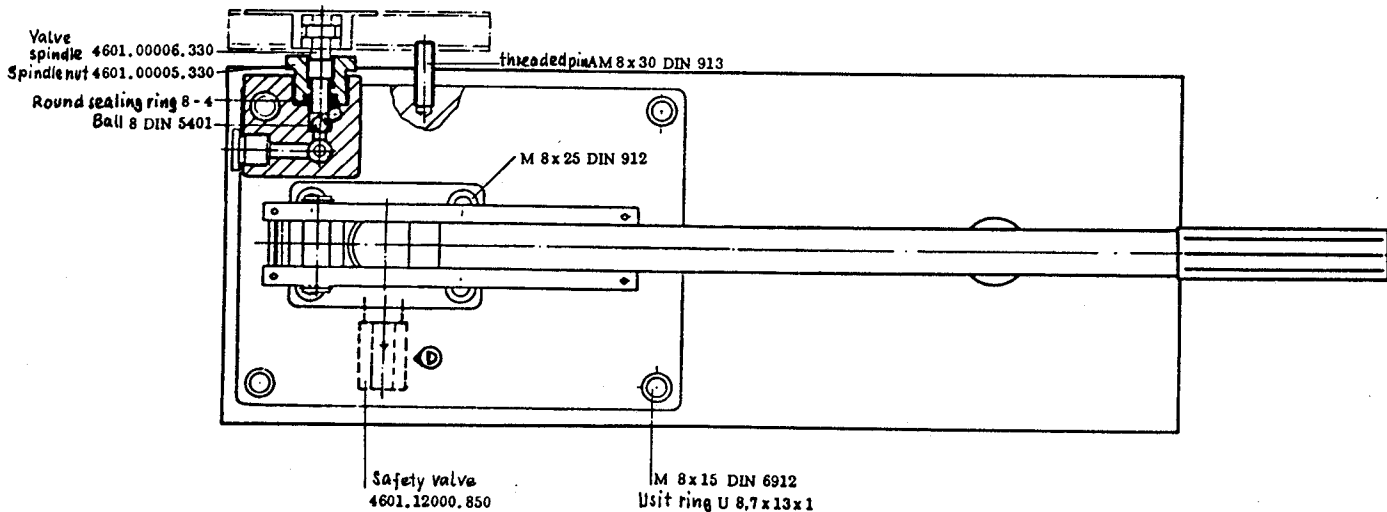
Spring clip 4601.10001.850

round sealing ring
R 170-4

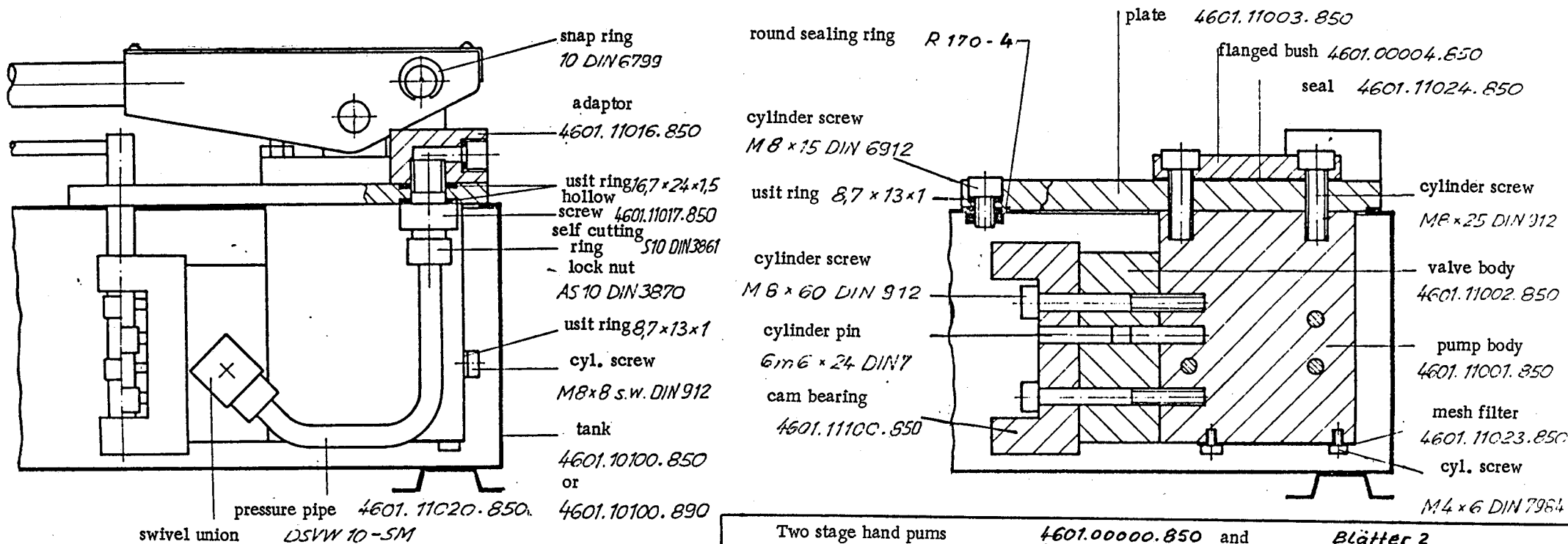
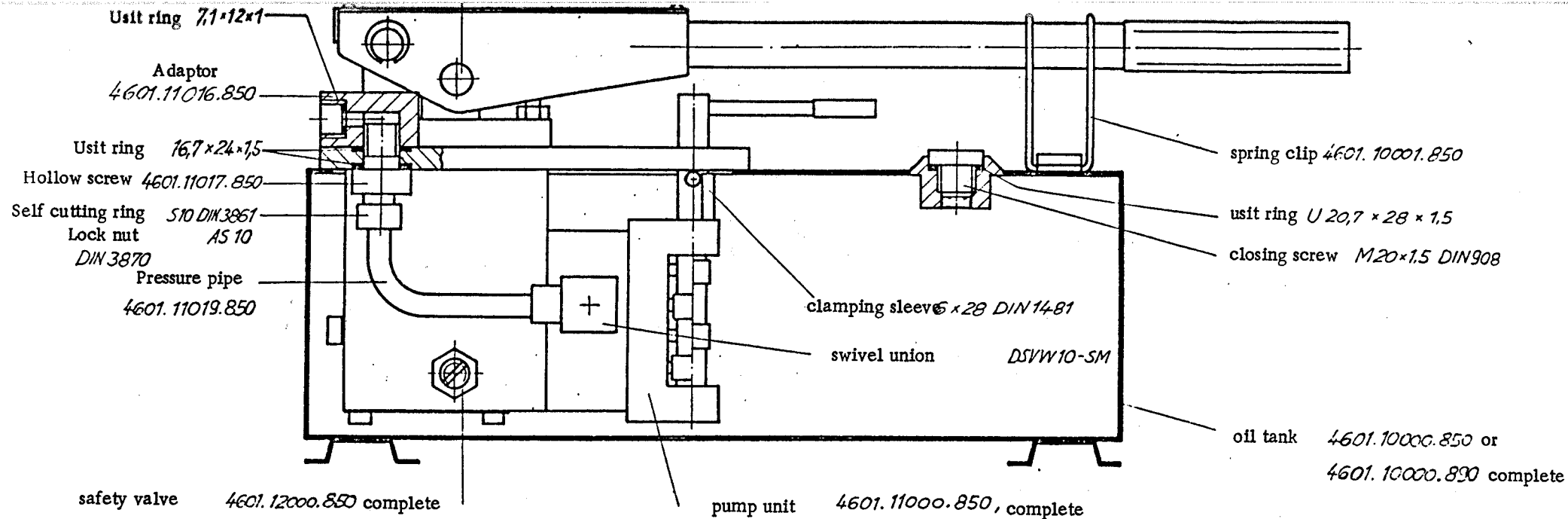
Tank 4601.10000.850 for 4601.00000.360
or 4601.10000.890 for 4601.00000.370



- A Suction valve
- B Pressure valve
- C Reversing valve
- D Safety valve



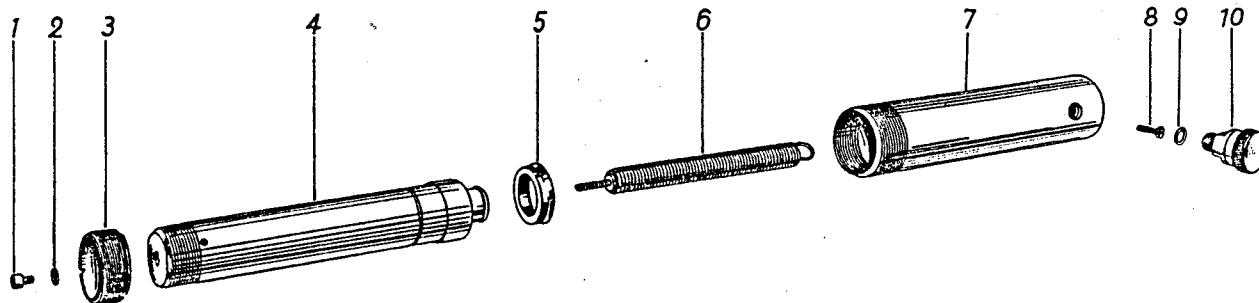
Spare part drawing
Hand pump 4601.00000.360 and 4601.00000.370
+ 4601.00000.380



Two stage hand pums

 4601.00000.850 and
 4601.00000.890

 Blätter 2
 Blatt 1



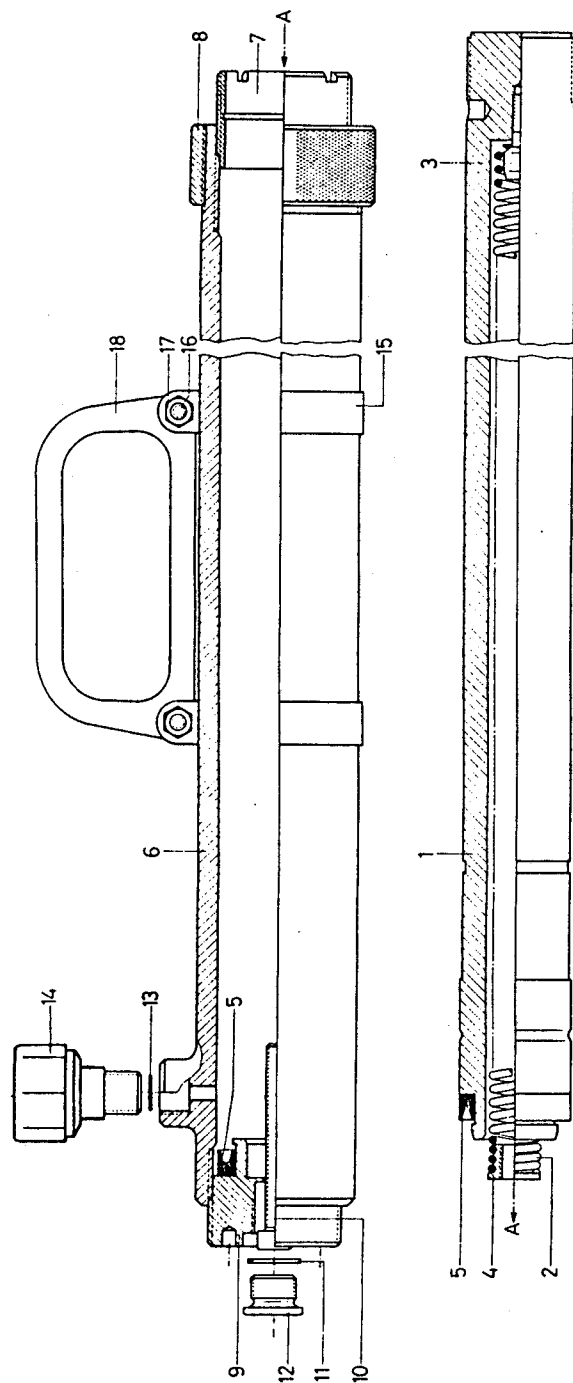
Mounting piston (4) and return spring (6):

Screw setscrew of return spring (6) by two turn only into assembled piston (4). Insert piston into pressure cylinder (7), fix spring eyelet with screw (8).

Turn setscrew of return spring (6) from outside anti-clockwise to run it down. Leave a little play between piston (4) and spring (5), so that piston can still be turned. Seal off tapped bore in piston (4) with screw (1) and sealing ring (2).

Tools: hexagon key, DIN 911, various sizes.

Item	Part Description	Ram 38 kN 4610.00000.070 Parts No.	Ram 38 kN 4610.00000.080 Parts No.	Ram 38 kN 4610.00000.090 Parts No.	Ram 60 kN 4610.00000.140 Parts No.	Ram 60 kN 4610.00000.150 Parts No.	Ram 100 kN 4610.00000.270 Parts No.	Ram 100 kN 4610.00000.280 Parts No.	Ram 100 kN 4610.00000.290 Parts No.	Ram 160 kN 4610.00000.390 Parts No.	Ram 200 kN 4610.00000.450 Parts No.	Ram 400 kN 4610.00000.610 Parts No.
1	Screw	4601.00009.220	4601.00009.220	4601.00009.220	4610.00005.150	4610.00005.150	4610.00006.290	4610.00006.290	4610.00006.290	4610.00006.290	Closing screw M 12x1,5 DIN 908	Closing screw M 16x1,5 DIN 908
2	Sealing ring	Sealing ring A 8 x 11,5 DIN 7603-Cu	Sealing ring A 8 x 11,5 DIN 7603-Cu	Sealing ring A 8 x 11,5 DIN 7603-Cu	Usit ring U 5,7 x 9 x 1	Usit ring U 5,7 x 9 x 1	Usit ring U 6,7 x 10 x 1	Usit ring U 6,7 x 10 x 1	Usit ring U 6,7 x 10 x 1	Usit ring U 6,7 x 10 x 1	Sealing ring A 12 x 16 DIN 7603-Cu	Sealing ring A 12 x 20 DIN 7603-Cu
3	Slotted nut	4610.00003.090	4610.00003.090	4610.00003.090	4610.00003.150	4610.00003.150	4610.00003.270	4610.00003.290	4610.00003.290	4610.00003.390	4610.00003.450	4610.00004.610
4	Piston	4610.00002.070	4610.00002.080	4610.00002.090	4610.00002.140	4610.00002.150	4610.00002.270	4610.00002.280	4610.00002.290	4610.00002.390	4610.00002.450	4610.00002.610
5	Grooved ring	VN20-47, 8V/504 20 Øx28 Øx8 x 5	VN20-47, 8V/504 20 Øx28 Øx8 x 5	VN20-47, 8V/504 20 Øx28 Øx8 x 5	N20-1, 14Pa/108 20 Øx36 Øx8 x 4	N20-1, 14Pa/108 20 Øx36 Øx8 x 4	N30-11, 14Pa/108 30 Øx45 Øx10 x 5	VN35-25, 8U/504 35 Øx45 Øx8 x 5	N30-11, 14Pa/108 30 Øx45 Øx10 x 5	NA57-11, 14Pa/108 57 Øx41 Øx10,5x12x7	N 45-6, 14Pa/108 45 Øx65 Øx10 x 5	N70-1, 14Pa/108 70 Øx90 Øx10 x 5
6	Return spring with setscrew	4610.00300.070	4610.00300.080	4610.00300.090	4610.00300.140	4610.00300.150	4610.00300.270	4610.00300.280	4610.00300.290	4610.00300.390	4610.00300.450	4610.00300.610
7	Pressure cylinder	4610.00001.070	4610.00001.080	4610.00001.090	4610.00001.140	4610.00001.150	4610.00001.270	4610.00001.280	4610.00001.290	4610.00001.390	4610.00001.450	4610.00001.610
8	Screw	Cyl. screw M 5 x 16 DIN 912	Cyl. Screw M 5 x 16 DIN 912	Cyl. Screw M 5 x 16 DIN 912	Cyl. Screw M 5 x 20 DIN 912	Cyl. Screw M 5 x 20 DIN 912	4610.00005.290	4610.00005.290	4610.00005.290	4610.00005.390	4610.00005.450	4610.00007.610
9	Washer	Washer A 10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10 x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu	Washer A10x13,5 DIN 7603-Cu
10	Coupler	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540
11	Protectice nut	4610.00004.090	4610.00004.090	4610.00004.090	4610.00006.150	4610.00006.150	4610.00007.290	4610.00007.290	4610.00007.290	4610.00006.390	4610.00006.450	4610.00006.610
12	Screw opposite coupler											Closing Screw M 16x1,5 DIN 908
13	Washer opposite coupler											Sealing ring A 16 x 20, DIN 7603-Cu
14	Grip (on cylinder)											4610.00005.610
15	Screw for grip (2)											Cyl. screw M10x1,5 DIN 912

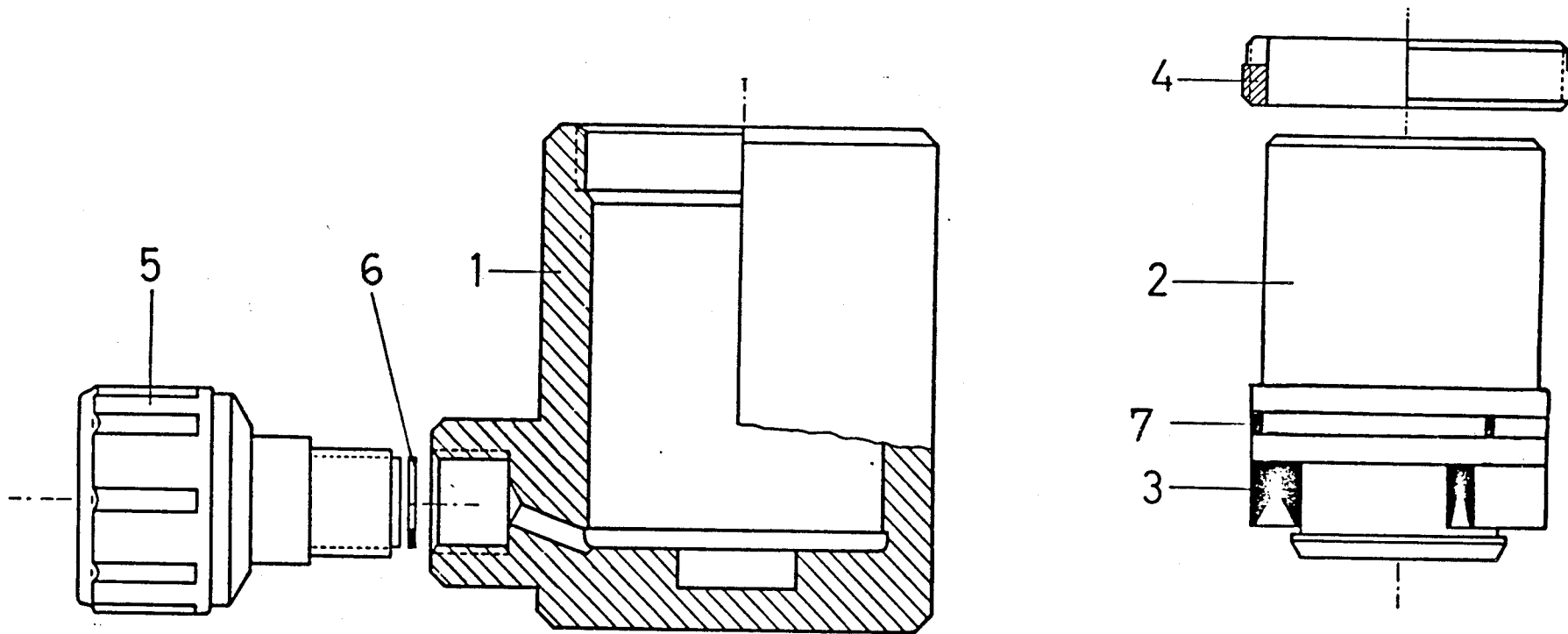


Mounting piston (1) and return spring (2)

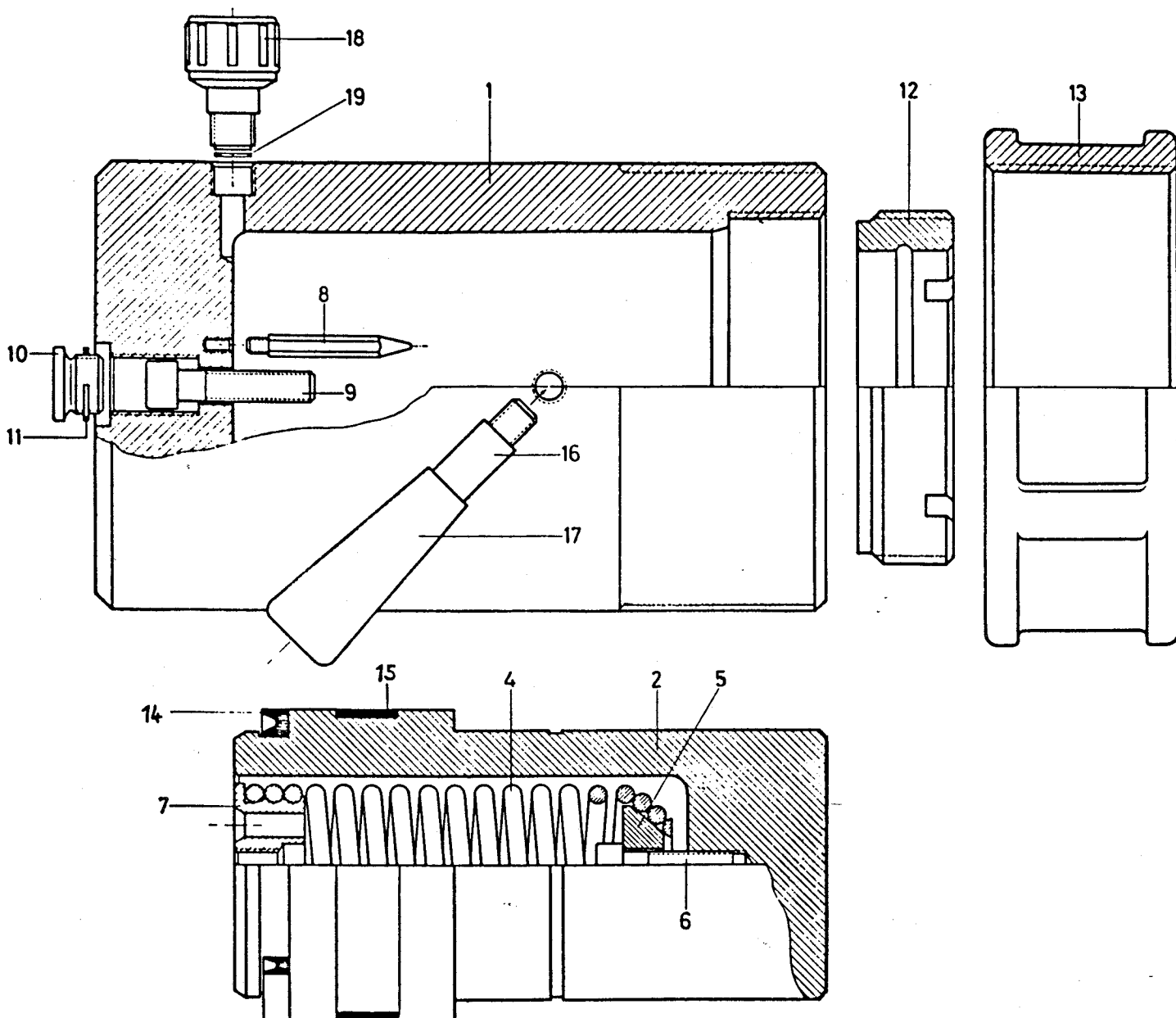
Turn setscrew (3) of return spring (2) into assembled piston (1) to the stop. Make sure that return spring can still be turned. Then screw spring holder (4) into return spring (2) and insert piston (1) into cylinder (6). Screw end plug (9) - possibly with screwed-in dowel which should be fitted through bore of spring holder - into cylinder (6). Insert cyl. screw (10) through centre bore of end plug (9), run it into spring holder (4) to pre-load return spring (12). Finally, tightly seal off centre bore in plug (9) with screw (12) and washer (11).

Tools: hexagon key, DIN 911, in various sizes.

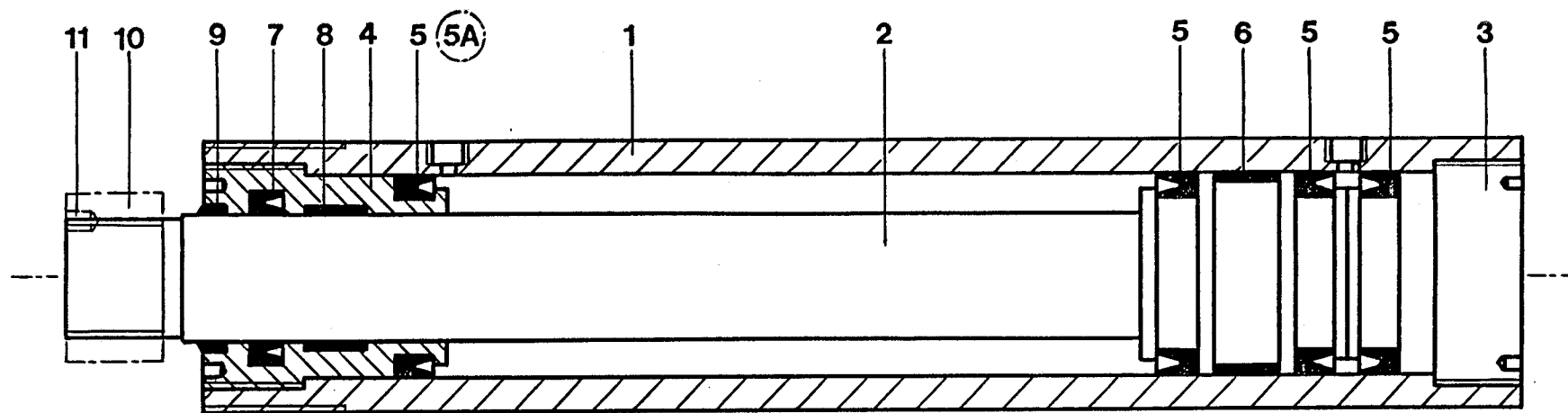
Item	Part Description	Ram 100 kN 4610.00000.310 Parts No.	Ram 200 kN 4610.00000.470 Parts No.	Ram 400 kN 4610.00000.620 Parts No.	Ram 650 kN 4610.00000.720 Parts No.	Ram 650 kN 4610.00000.730 Parts No.	Ram 1000 kN 4610.00000.850 Parts No.	Ram 1000 kN 4610.00000.860 Parts No.
1	Piston	4610.00001.310	4610.00001.470	4610.00002.620	4610.00002.720	4610.00002.730	4610.00002.850	4610.00002.860
2	Return spring	4610.00004.310	4610.00004.470	4610.00005.620	4610.00005.720	4610.00004.730	4610.00004.850	4610.00004.860
3	Setscrew Spring plate	4610.00200.310	4610.00200.470	4610.00200.620	4610.00009.710 4610.00004.710	4610.00009.710 4610.00004.710	4610.00009.710 4610.00004.710	4610.00009.710 4610.00004.710
4	Spring holder	4610.00002.310	4610.00002.470	4610.00003.620	4610.00003.720	4610.00003.720	4610.00003.720	4610.00003.720
5	Grooved ring	N30-11, 14Pa/108 30 Ø x 45 Ø x 10 x 5	N45-6, 14Pa/108 45 Ø x 65 Ø x 10 x 5	N70-1, 14Pa/108 70 Ø x 90 Ø x 10 x 5	N95-4, 14Pa/108 95 Ø x 115 Ø x 10 x 5	N95-4, 14Pa/108 95 Ø x 115 Ø x 10 x 5	N125-2, 14Pa/108 125 Ø x 145 Ø x 10 x 5	N125-2, 14Pa/108 125 Ø x 145 Ø x 10 x 5
6	Pressure cylinder	4610.00100.310	4610.00100.470	4610.00001.620	4610.00000.720	4610.00001.730	4610.00001.850	4610.00001.860
7	Screw	4610.00003.310	4610.00003.470	4610.00004.620	4610.00004.720	4610.00003.730	4610.00003.850	4610.00003.860
8	Protective nut	4610.00007.290	4610.00006.450	4610.00008.610	4610.00011.710	4610.00011.710	4610.00005.840	4610.00005.840
9	End plug Pin (screwed in)	4610.00005.310	4610.00005.470	4610.00005.600	4610.00006.720 4610.00009.720	4610.00006.720 4610.00009.720	4610.00005.850 4610.00009.729	4610.00005.850 4610.00009.720
10	Cyl. Screw	4610.00006.310	4610.00006.470	4610.00006.620	4610.00007.720	4610.00005.720	4610.00007.850	4610.00005.860
11	Sealing ring	Sealing ring A 16 x 20 DIN 7603-Cu	Sealing ring A 20 x 24 DIN 7601-Cu	Sealing ring A 20 x 24 DIN 7603-Cu	Sealing ring A 22 x 27 DIN 7603-Cu	Sealing ring A 22 x 27 DIN 7603-Cu	Sealing ring A 22 x 27 DIN 7603-Cu	Sealing ring A 22 x 27 DIN 7603-Cu
12	Screw	End plug M16x1,5 DIN 908	End plug M20x1,5 DIN 908	End plug M20x1,5 DIN 908	End plug M22x1,5 DIN 908	End plug M22x1,5 DIN 908	End plug M22x1,5 DIN 908	End plug M22x1,5 DIN 908
13	Sealing ring	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu	Sealing ring A 10 x 13,5 DIN 7603-Cu
14	Coupler	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540
15	Clips (2)	4610.00008.310	4610.00007.470	4610.00008.620	4610.00008.720	4610.00008.720	4610.00008.850	4610.00008.850
16	Hexagon screw	Hexagon screw M 6 x 30 DIN 931	Hexagon screw M 6 x 30 DIN 931	Hexagon screw M 12x45 DIN 931	Hexagon screw M 12x45 DIN 931	Hexagon screw M 12x45 DIN 931	Hexagon screw M 12x45 DIN 931	Hexagon screw M 12x45 DIN 931
17	Hexagon nut	Hexagon nut M 6 DIN 934	Hexagon nut M 6 DIN 934	Hexagon nut M 12 DIN 934	Hexagon nut M 12 DIN 934	Hexagon nut M 12 DIN 934	Hexagon nut M 12 DIN 934	Hexagon nut M 12 DIN 934
18	Grip	4610.00007.310	4610.00007.310	4610.00007.620	4610.00007.620	4610.00007.620	4610.00007.620	4610.00007.620



item	description	ram 32 kN 4610.00000.051 order No.	ram 60 kN 4610.00000.110 order No.	ram 100 kN 4610.00000.265 order No.	ram 200 kN 4610.00000.425 order No.	ram 200 kN 4610.00000.445 order No.	ram 400 kN 4610.00000.585 order No.	ram 400 kN 4610.00000.595 order No.
1	pressure cylinder	4610.00001.051	4610.00100.110	4610.00100.260	4610.00100.420	4610.00100.440	4610.00001.580	4610.00001.590
2	piston	4610.00002.051	4610.00001.110	4610.00001.265	4610.00001.425	4610.00001.445	4610.00002.585	4610.00002.595
3	grooved ring	VN 10-29,8U/504 10øx26øx8x4	VN 20-51 20/36 ø x 8	NA 400 - 45/30 ø x 10	UBA 65/45 Øx 10	UBA 65/45 Øx 10	UBA 90/75 Øx 12	UBA 90/75 Øx 12
4	slotted nut	4610.00003.051	4610.00002.110	4610.00003.270	4610.00002.420	4610.00002.440	4610.00003.580	4610.00003.580
5	coupler	4606.11000.540 dazwischen: Schlauch und 4610.00004.050	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540	4606.11000.540
6	Usit ring	Usitring U 7,1 x 12 x 1	A 10 x13,5 DIN 7603 -Cu	Dichtring A 10x13,5 DIN 7603-Cu	Dichtring A 10x13,5 DIN 7603-Cu	Dichtring A10x13,5 DIN 7603-Cu	Dichtring A10x13,5 DIN 7603-Cu	Dichtring A10x13,5 DIN 7603-Cu
7	piston guide ring			S 50703-0450-A Turcite-B	EKF 65	EKF 65	EKF 90	EKF 90



item	description	ram 650 kN 4610.00000.715 order No.	ram 1000 kN 4610.00000.845 order No.
1	press cylinder	4610.00001.710	4610.00001.840
2	piston	4610.00002.715	4610.00002.845
3			
4	return spring	4610.00007.710	4610.00004.840
5	spring plate	4610.00004.710	4610.00004.710
6	set screw	4610.00009.710	4610.00009.710
7	spring holder	4610.00003.720	4610.00003.720
8	hexagon pin	4610.00010.710	4610.00010.710
9	cyl. screw	4610.00008.710	4610.00008.710
10	end plug	M 22x1,5 DIN 908	M22x1,5 DIN 908
11	sealing ring	A 22x27 DIN 7603-Cu	A 22 x27 DIN 7603-Cu
12	slotted screw	4610.00003.710	4610.00003.810
13	protective nut	4610.00011.710	4610.00005.840
14	gr. ring	UBA 115/95 ϕ x 12 - 008 821 149 825	UBA145/125 ϕ x15-008821149820
15	piston guide ring	EKF 115-008 821 270 230	EKF 150 - 008821270235
16	handle (1 pair)	4610.00005.711	4610.00005.711
17	taper grip	32 x 16 NQ 92	32 x 16 NQ 92
18	coupler	4606.11000.540	4606.11000.540
19	sealing ring	A 10x13,5 DIN 7603-Cu	A 10x13,5 DIN 7603-Cu



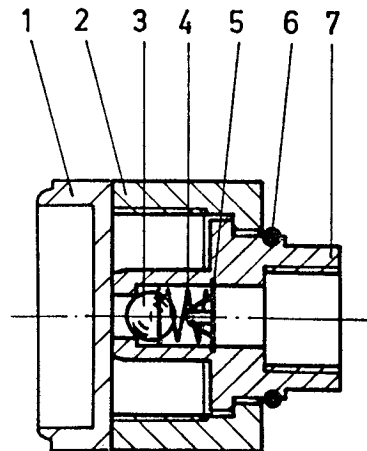
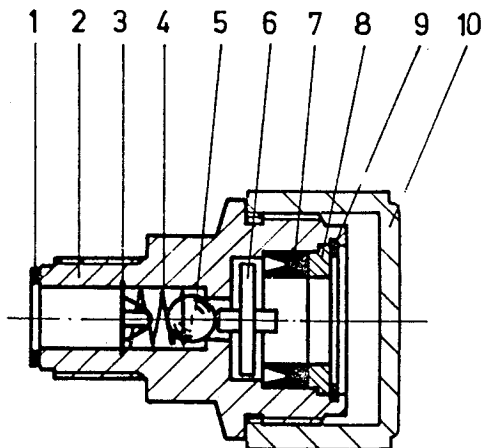
push / pull rams (double acting rams)

item	description	100/160 4615.00000.015 order No.	100/320 4615.00000.155 order No.	270/200 4615.00000.115 order No.	270/400 4615.00000.025 order No.	400/200 4615.00000.035 order No.	400/400 4615.00000.135 order No.	650/200 4615.00000.045 order No.	650/400 4615.00000.075 order No.	1000/200 4615.00000.085 order No.	1000/400 4615.00000.095 order No.
1	press calinder	4615.00100.015	4615.00100.155	4615.00100.115	4615.00100.025	4615.00001.035	4615.00001.135	4615.00001.045	4615.00001.075	4615.00001.085	4615.00001.095
2	piston with rod	4615.00001.015	4615.00001.155	4615.00001.115	4615.00001.025	4615.00002.035	4615.00002.135	4615.00002.045	4615.00002.075	4615.00002.085	4615.00002.095
3	hollow screw	4615.00003.015	4615.00003.015	4615.00002.115	4615.00002.115	4615.00003.035	4615.00003.035	4615.00003.045	4615.00003.045	4615.00003.085	4615.00003.085
4	cover			4615.00003.115	4615.00003.115	4615.00004.035	4615.00004.035	4615.00004.045	4615.00004.045	4615.00004.085	4615.00004.085
5	grooved ring	NA 400 - 45/30 ϕ x 10	NA 400 - 45/30 ϕ x 10	UBA - 45/60 ϕ x 12	UBA - 75/60 ϕ x 12	UBA - 90/75 ϕ x 12	UBA - 90/75 ϕ x 12	UBA - 115/95 ϕ x 12	UBA - 115/95 ϕ x 12	UBA - 145/125 ϕ x 15	UBA - 145/125 ϕ x 15
6	piston guide ring	EKF 45	EKF 45	EKF 75	EKF 75	EKF 90	EKF 90	EKF 115	EKF 115	EKF 150	EKF 150
7	grooved ring	UB 32/40 ϕ x 8	UB 32/40 ϕ x 8	UBJ - 45/60 ϕ x 10	UBJ - 45/60 ϕ x 10	UBJ - 65/80 ϕ x 12	UBJ - 65/80 ϕ x 12	UBJ - 75/95 ϕ x 12	UBJ - 75/95 ϕ x 12	UBJ - 110/130 ϕ x 15	UBJ - 110/130 ϕ x 15
8	piston rod guide ring	FB - 10 x 2 x 104	FB - 10 x 2 x 104	FB - 20 x 2 x 145	FB - 20 x 2 x 145	FB - 20 x 2 x 206	FB - 20 x 2 x 206	FB - 25 x 2 x 239	FB - 25 x 2 x 239	FB - 35 x 2 x 348	FB - 35 x 2 x 348
9	wiper	P 7 - 32/45 ϕ x5x7	P 7 - 32/45 ϕ x5x7	UAS - 45/55 ϕ x7x10	UAS - 45/55 ϕ x7x10	UAS - 65/75 ϕ x7x10	UAS - 65/75 ϕ x7x10	UAS - 75/85 ϕ x7x10	UAS - 75/85 ϕ x7x10	UAS - 110/120 ϕ x7x10	UAS - 110/120 ϕ x7x10
10	threaded nipple	4610.00004.010	4610.00004.010								
11	threaded pin	AM 6 x 8 DIN 914	AM 6 x 8 DIN 914								
5A	round seal	R 38 - 3	R 38 - 3								

Coupling, new version

Coupler 4606.11000.540
for ram

Coupler 4606.12000.540
for hose



Coupler No. 4606.11000.540

Item	Part Description	Parts No.
1	Sealing ring	A 10 x 13,5 DIN 7603-Cu
2	Coupler housing	4606.11001.540
3	Spring holder	4606.11003.540
4	Spring 7 x 10 x 0,8 x 55 NQ 90	9932.00042.550
5	Ball	7 II DIN 5401
6	Plunger plate	4606.11002.540
7	Grooved ring	VN 12 - 47, 8V/504 (12 Ø x 18 Ø x 6 x 4)
8	Spacer ring	4606.11002.500
9	Snap ring	20 x 1 DIN 472
10	End cap	4606.11007.501

Coupler No. 4606.12000.540

Item	Part Description	Parts No.
1	End cap	4606.12003.501
2	Locknut	4606.12002.540
3	Ball	7 II DIN 5401
4	Spring 7 x 10 x 0,8 x 5,5 NQ 90	9932.00042.550
5	Spring holder	4606.11003.540
6	Snap ring	22 DIN 9045
7	Adaptor	4606.12001.540